

GAS PROCESSING & METHANOL COMPLEX PROJECT NO. 11998A

Project	FEED for Gas Gathering Pipelines & Associated Infrastructure
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Materials Specification and Corrosion Management

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Revision History

Rev No.	Date	Description of Revision
R01	13-07-22	Issued for Review

Revision Philosophy:

- a. All documents for review shall be issued at R01, with subsequent R02, R03, etc as required.
- b. All documents approved for issue or approved for design shall be issued at A01 with subsequent A02, A03, etc. as required. (Management of Change is required for A02, A03, etc.)
for the subsequent project phases, post FEED, it is expected that:
- c. All Detailed Design documents for review shall be issued at D01, with subsequent D02, D03, etc. as required.
- d. All documents approved for construction shall be issued at C01 with subsequent C02, C03, etc. as required. (Management of Change is required for C02, C03, etc.)
- e. All approved "As-Built" documents shall be issued at Z01, with subsequent Z02, Z03, etc as required. (Use versions Z01.1, Z01.2, Z01.3, etc to review the "As-Built" document to Z02).

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SUMMARY PAGE

Client:	Brass Fertilizer & Petrochemical Company Limited	
Project:	Gas Processing & Methanol Complex	Project No. 11998A
Facility:	Gas Gathering Pipelines & Associated Infrastructure	Doc No. EA-BFPCL-GPMC-GGPL-FEED-MCM-DOC-001

The administration of this document is in respect of FEED for Gas Gathering Pipelines & Associated Infrastructure for the Gas Processing & Methanol Plant Project of BFPCL.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2. The focus of the present FEED is on Phase 1, while Phase 2 will be implemented later.

It is also a precursor and accompaniment to the FEED study that will lead to the project's Financial Closure and implementation.

The report has relied on and used data and reference documents provided by the client, and if further documents and information become available, Epic Atlantic reserves the right to update the opinion set out in this report.

For better appreciation, this document should be read in conjunction with the references listed below, which are the Concept Select Study reports that constitute basis for the Conceptual Study.

KEYWORDS: BASIS OF DESIGN, PIPELINE, MANIFOLD, PLATFORM, MULTIPHASE FLOW, ROW, PTS

REFERENCES

Document No.	Description
EA/SPDC-BFPC UGS-GEP/SVY-RPT-001	Pipeline Route Selection Study
	Upstream Gas Gathering Concept Select & Revalidation Report
EA-BFPCL-GPMC-GGPL-FEED-GEN-RPT-002	Basis of Design
EA-BFPCL-GPMC-GGPL-FEED-GEN-RPT-001	Design & Operating Philosophy
EA/SPDC-BFPC UGS-GEP/CSM-RPT-002	Brone Topographical & Geotechnical Survey
	FISAS Draft ESIA Report (November 2020)

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Abbreviations

Abbreviation	Meaning
BEPCL	Brass Fertilizer & Petrochemical Company Limited
ANSI	American National Standards Institutes
API	American Petroleum Institute
ASME	American Society of Mechanical Engineers
ASTM	American Society for Testing and Materials
BBOU	Bubouwe
BE	Bevel end
BBE	Bevel Both Ends
BFW	Bevel for Welding
BLE	Bevel Large End
CP	Cathodic Protection
DN	Diameter
EP	Epic Atlantic
ELEP	Elepa
ERW	Electrical Resistance Welding
H ₂ S	Water
ID	Internal Diameter
OD	Outer Diameter
ISO	International Standard Organisation
LSAW	Longitudinal Submerged Arc Welded
MS	Mild Steel
NACE	National Association of Corrosion Engineers
NPS	Nominal Pipe Size
OCTG	Oil Country Tubular Goods
pCO ₂	Partial Carbon Dioxide Pressure
PHWT	Post Heat Weld Treatment
SA	Sacrificial Anodes
SPDC	Shell Petroleum Development Company
SSAW	Spiral Submerged Arc Welding
STD	Standard Weight Wall Thickness

Abbreviation	Meaning
TEC	Tata Engineering Company Limited
XS	Extra Strong Wall Thickness
XX	Double Extra Strong Wall Thickness

1. Introduction

1.1. Background Information

Brass Fertilizer and Petrochemical Company Limited (BFPCL) is developing a 10,000 metric tonnes per day (MTPD) capacity greenfield methanol production facility in Akabel (Odioma), Brass LGA, Bayelsa state. The plant complex is to be based on natural gas feedstock from 5 SPDC fields in OMLs 33, 35 & 36.

The Project involves the development, construction, and operation of a plant comprising:

- Two trains of 5,000 MTPD methanol plant, producing a total of 10,000 MTPD of methanol
- 340 MMscfd Gas Processing Plant ("GPP"), which will extract NGL from the natural gas feedstock, prior to feeding the lean gas to the methanol plant.
- Gas gathering pipelines and associated manifolds transporting the natural gas from five fields, viz: Bubouwe Bou (BBOU), Elepa (ELEP), Nun River (NUNR), Seibou (SEIB), and Opugbene (OPUG) to the GPP at Akabel.
- Product Export Lines and SBM for the export of the methanol, and NGL of the GPP.
- All offsite, utility, and off-plot facilities (including product storage, internal power generation, steam generation, and export jetty) for the operation of the processing plants.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2.

The focus of the present FEED is on Phase 1 as shown in figure 1 below, while Phase 2 will be implemented later.

TCE is the Project Management Consultant for the project.

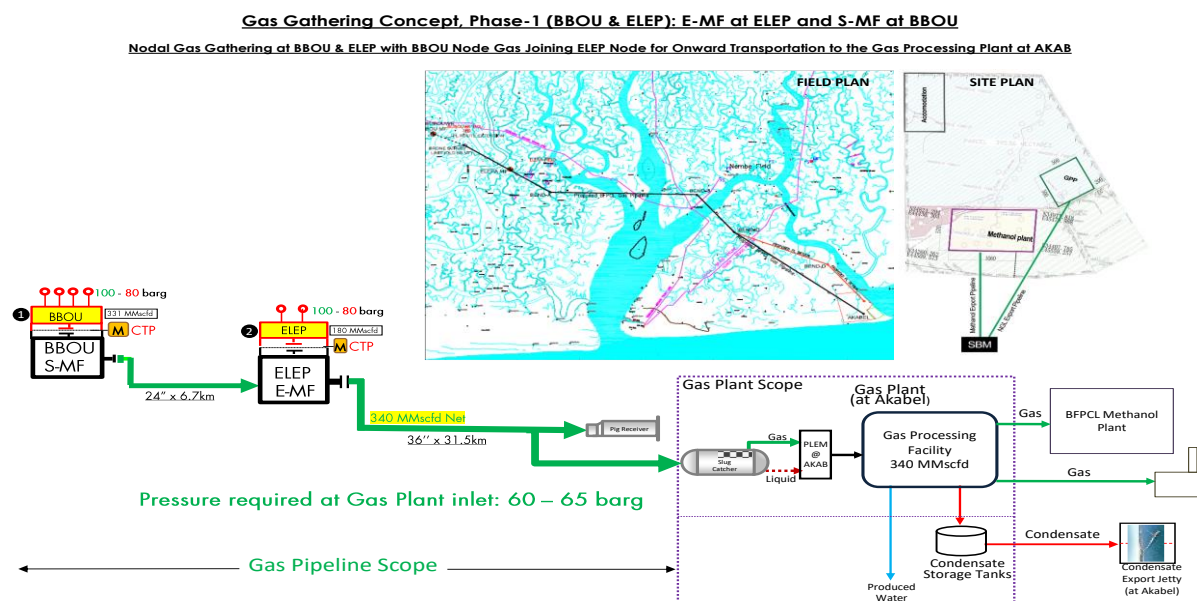


Figure 1.1: Gas Gathering Concept, Phase 1

1.2. Purpose

The purpose of this document is three-fold:

1. Identification of best pipeline materials for each of the line segments at optimised life cycle costs, utilising best industry practice.
2. Assessment of the risks associated with the specific selection.
3. Specification of the corrosion management system that will mitigate the risk.

The objective is the optimal specification of materials and the relational corrosion management systems that can contend with the process & environmental conditions, and operational requirements of the pipeline system

The corrosion barriers and control systems allow for the monitoring, maintenance and management review for continuous improvement.

The FEED shall fully comply with all specified relevant contractual requirements.

3. Regulations, Codes And Standards

Document Number	Document Title
DEP 31.38.01.31-Gen	Shop and Field Fabrication of Piping
DEP 02.00.00.10-Gen	Preparation of Technical Drawings
DEP 74.00.10.10-Gen	Shop and Field Pressure Testing of Piping and Process Systems
DEP 31.38.01.29-Gen	Pipe Supports
DEP 30.00.60.20-Gen	Human Factors Engineering – Physical Workspace Design
DEP 31.40.20.37-Gen.	Line Pipe for Critical Service (Amendments/Supplements to ISO 3183:2007)
DEP 31.40.00.10-Gen.	Pipeline engineering (amendments/supplements to ISO13623)
DEP 61.40.20.30-Gen	Welding of pipelines and related facilities (amendments/supplements to ANSI/API STD 1104)
DEP 31.40.20.33-Gen	Line pipe induction bends (amendments/supplements to ISO 15590-1)
DEP 30.10.02.16-Gen	Hydrogen induced cracking sensitivity test (amendments/supplements to NACE TM0284)
DEP 82.00.10.10-Gen	Project Quality Assurance
DEP 30.10.02.11-CSPC	Metallic Materials – Selected Standards
DEP 30.10.73.10-CSPC	Cathodic Protection
DEP 30.10.73.31-CSPC	Design of Cathodic Protection Systems for Onshore Buried Pipelines
DEP 30.48.00.31-CSPC	Painting and Coating of New Equipment
DEP 31.40.00.10-CSPC	Pipeline Engineering
ASME B16.5	Pipe Flanges and Flanged Fittings
ASME B16.10	Face to Face and End to End Dimensions of Valves
ASME B16.20	Metallic Gasket for Pipe Flanges, Ring Joints, Spiral Wound and Jacketed.
ASME B16.21	Non-Metallic Flat Gaskets for Flanges
ASME B16.25	Butt Welding Ends
ASME B16.34	Valve – Flanged, Treaded, and Welding End
ASME B16.47	Large Diameter Steel Flanges NPS 26 through NPS 60
ASME B31.8	Transportation of Petroleum Gas
ASTM A53	Hot Rolled Carbon Steel Pipe
ASTM A105	Forgings, Carbon Steel for Piping Components

Document Number	Document Title
ASTM A106	Seamless Carbon Steel Pipe for High Temperature Service
ASTM A182	Forged or Rolled Alloy-Steel Pipe Flanges, Forged Fittings, and Valves and Parts for High-Temperature Services
ASTM A370	Standard Test Methods and Definitions for Mechanical Testing of Steel Products
API 5L	Carbon Steel Line Pipes
API 5LW	Transportation of Line Pipe
API 5L7	Unprimed Internal Fusion Bonded Epoxy Coating of Line Pipe.
API 5LC	Duplex Stainless Steel Pipe

The latest versions of the regulations, codes, and standards and extant amendments or supplements shall apply. Local laws regulating the oil and gas industry in Nigeria shall be strictly adhered to. The order of precedence for the documents shall be as follows:

- Nigerian Local Standards and Codes
- The Contract
- Project Specifications
- Employer's Standards
- Shell DEPs
- International Codes and Standards

In all cases, the most stringent shall apply except where constraints dictate otherwise.

Conflicts between specifications and referenced Codes and Standards or deviations thereof shall be referred in a Technical Query (TQ) to the EMPLOYER for resolution.

4. Corrosion Analysis, Protection & Control Management Philosophy

This document is the corrosion analysis and corrosion protection & control management philosophy adopted for the pipelines from the Bubouwe manifold to Akabel in Brass Island, Bayelsa State. It stems down from the conceptual design. It is the phase 1 pipelines' FEED development implementation for Brass Fertilizer Petrochemical and Company Limited (BFPCL). It sets out the basic principles to shape the processes that capture a total corrosion management flow from the FEED, allowing for the corrosion barriers introduced to be monitored, maintained and undertaking the required management review for continuous improvement of the system as shown in the corrosion process flow in Fig 3.1 below. Fig 3.1 reflects the inputs from the process and the relevant management of the corrosion actions.

In essence, the containment of the hydrocarbon and technical/operational integrity within the pipelines (Fittings, Valves, and Ancillary Equipment) is the life time goal at optimal cost. Hence, all Failure Mode Analysis and Effects (FMAE) are thoroughly appraised at this FEED and followed up to the abandonment. This should be done at affordable rates.

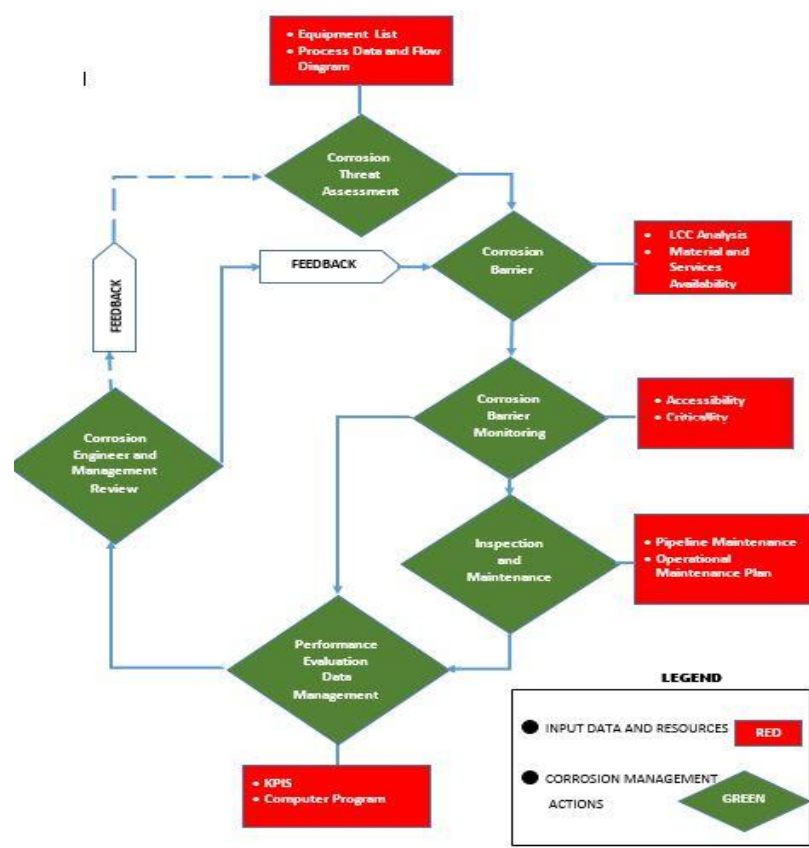


Fig 3.1. Process Flow of Corrosion Analysis, Corrosion Protection and Management Philosophy

4.1. General

This philosophy is in accordance with the NACE provision and approach as shown in the flow process in fig 3.1 above.

This sets out in the following order:

- Identifying the Threats
 - Internal Threats
 - External Threats
- Prediction and Assessment of the Corrosion Damage
- Corrosion Control
- Management of Corrosion and Control Strategies.
- Corrosion Economics

4.1.1. Identification of the Corrosion Threats

The threats or failure mechanism are from the products being transported (Internal Corrosion) or the surrounding environment of the outside surface of the pipeline (external corrosion). Consequently, there are two broad areas of threats on this project:

- Internal Corrosion
- External Corrosion

4.1.1.1. Internal Threats

Input Information and data have been analysed for the internal threats to the pipeline materials. These included identifying from the fluid composition the main corrosion active species from:

- H₂S - sour corrosion
- CO₂ - sweet corrosion
- O₂ - oxygen corrosion
- Acid corrosion (organic/inorganic)
- Microbiological corrosion

4.1.1.2. External Threats

These are the threats posed by the external environment to the pipelines and equipment.

4.1.2. Prediction of Corrosion and Assessment

Once the corrodants or corrosion species are identified, the next action is to carry out the corrosion analysis utilising industry-approved computer programmed tools like Hydrocor, or De-Waard and Milliams nomogram for CO₂, H₂O and CO₂+H₂S predictions of the corrosion rate. These rates are extrapolated to give the corrosion allowance in terms of the design life. Secondly, the external corrosion is predicted using input data from the soil or submerged environmental resistivity data inputted from the resistivity survey. Again, splash zone area and

the atmospheric area are predetermined by experience and international classifications (NACE, ISO, DVN).

4.1.3. Corrosion Control

In this FEED we shall design the corrosion control by starting with selection of materials and/or establishing the corrosion barriers needed for the technical integrity of the facility. The philosophy for the pipeline material selection shall cover a wide range of materials (MS, low C-Mn alloy, CRA, Non – Ferrous Materials, Titanium, Composite Materials etc). And this will be selected at an optimal cost estimate and the best Life Cycle Cost (LCC) (Ref. Concept report Doc. No. EA SPDC – BFPC UGS MAT RPT 001 A01).

The external control barriers shall be achieved by design implementation of pipeline coatings and cathodic protection. Again, materials at splash zone shall be protected with splash zone treatment while above this zone shall be protected by painting.

4.1.4. The Strategies for Management of Corrosion and Control

This shall be defined by the quality of the implementation of the design process with relevant standards and codes and selection of materials. The strategies shall allow for proper maintainability, monitoring and continuous feedback into the system. The design must make provision for the requirement like pigging with the design for traps and chemical injection system using pumps & storage facilities for the inhibitor.

4.1.5. Corrosion Economics

Thermodynamically, metals are unstable and tend to revert to their natural stable state through corrosion process. This FEED philosophy on the economics of corrosion of steel is to achieve technical integrity of the pipeline system, while maintaining safety to personal and the environment, for the period of its design life. This is implemented through a combination of materials specification, anti-corrosion treatment, and processes at the best possible cost.

Three main economic breakdowns are considered here, with decisions based on Net Present Value (NPV):

- Analysing the essential cost components.
- The possibility of reducing significantly, the cost of anti-corrosion protection of the steel pipe, barriers, maintenance, and inspection throughout life time.
- The possibility of using new optional materials, subject to Life Cycle Cost (LCC) cost comparison and Total Cost of Ownership (TCO).

5. Materials Selection Philosophy

5.1. General

This materials selection philosophy document describes the processes and procedures that allow for the optimal selection of materials for this FEED phase1 project and the sub-components thereof for the:

24"x6.7km BBOU – ELEP Gas line and

36"x31.5km ELEP – AKABEL Gas Transmission Line

The process shall lead to the determination of the best relevant material after appropriate analysis and screening approach. The specifications and the procedures used shall cover

- The Line – Pipes
- The Piping
- The Equipment
- The Isolation Joints / Insulation Flanges
- (Thermal) Insulation (where relevant)

5.2. Purpose

The purpose is to establish procedures, specifications and approaches to contend with in the systematic choice of materials and their sub-components for this phase 1 FEED in accordance with international guidelines, codes & standards.

5.3. Pipe Material Types and Selection Processes

Pipe material selection of various components is dependent on the type of materials it transports. The relevant design codes and standards associated with this wet gas transportation are ASME-B31.1, B31.3, B31.8, ASTM-A53, A106, A234, A333, A671/2, API-5L and AWS.

Frequently, the groups used for line pipe and piping materials are:

- High Carbon steel – over 0.6% C in steel
- Medium Carbon steel - between 0.3% to 0.6% C in steel whilst
- Mild Steel is less than 0.3%. C.

For caution, carbon content above 0.35% makes the steel to become brittle and more than 0.43% is a challenge to welding. (Ref. carbon equivalent formula).

The line pipe and piping are graded according to strength (ASTM-A53, ASTM-A106).

- Grade C – Highest Strength
- Grade B – Less Strength
- Grade A – A106 is preferred at high temperature.

The cheap cost price of CS is what makes it useful for engineering applications, however the properties of API specifications 5L or ISO 3183 are better for line pipe production. Again, Carbon Manganese low alloy steel are now more favoured for line pipe materials production because of improved cost.

The processes of selection are now given in flowchart in Fig 4.2 below. The manganese including Nb, V and Ti further improve the precipitation, and strength properties by micro alloying of the CS. Hence, this composition is more preferred. Again, it is noted to have more improved quality for corrosion resistance (NACE MR0175 / ISO 15156-1). Some CO₂ corrosion improvement are also noted in comparison. For the purpose of this specification a 25% reduction may be accommodated on Hydrocor calculation at below 60 °C of operational temperatures (Ref, Analysis of the corrosion scales formed on API 5L X70 and X80 steel pipe in the presence of CO₂)

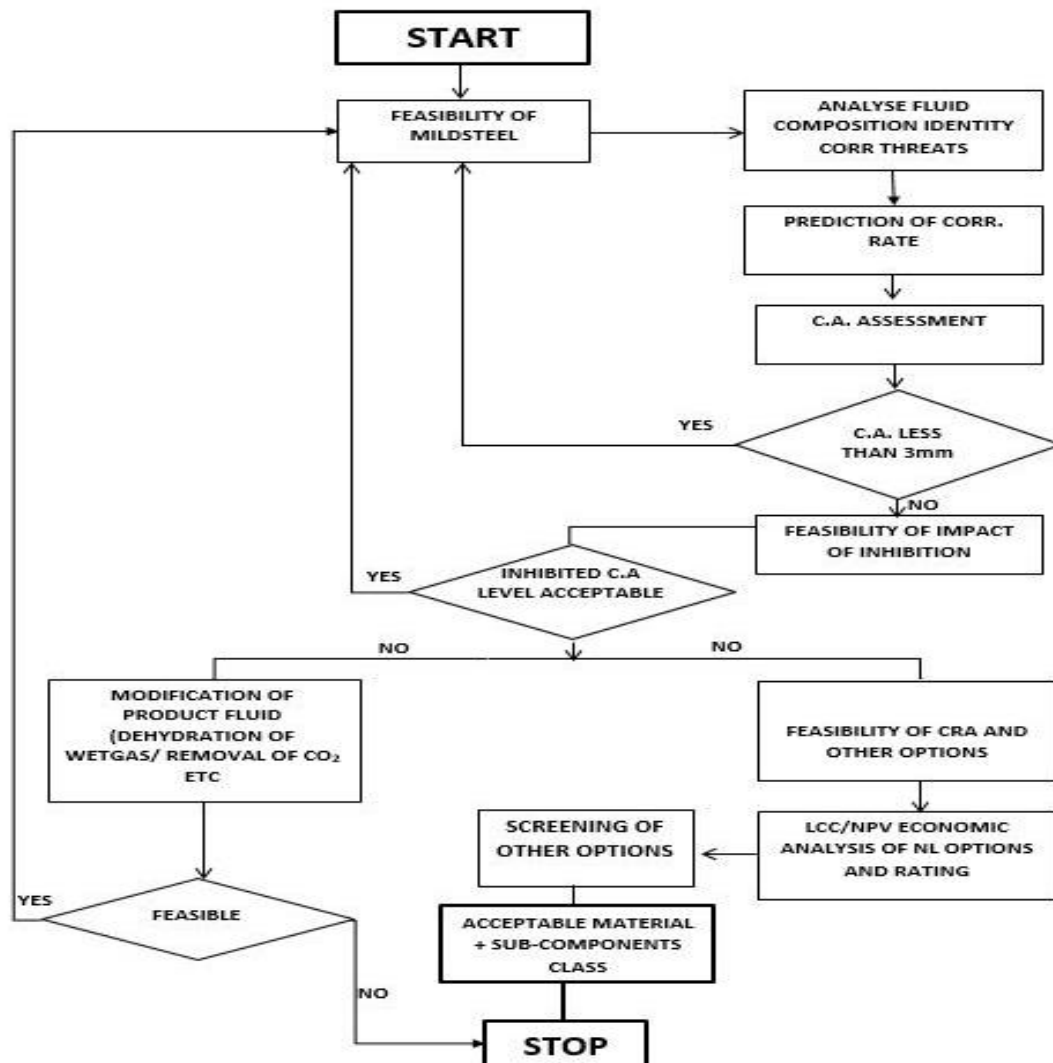


Fig 4.2: Process Steps for Materials Selection

The second part of this section is the selection processes as given in Fig 4.2 above. Three main decision areas are noted below:

- The outcome of the Hydrocor analysis is less than 3mm CA, - Mild or Low C-Mn alloy now become the material of choice. And the preferred CS is grade X70 because of its properties and lower cost price.
- If not, an inhibition process is reviewed at a 92.5% minimum efficiency without down time to inhibition pumping. This decision is based on acceptable level of CA.
- If not acceptable, optional analysis shall be reviewed and a selection shall be made accordingly.

5.4. Critical Parameters of Pipe Materials Selection

The critical parameters for pipe materials selection shall be reviewed in three discipline areas outlined below:

- The outcome of the hydraulic simulation studies (Process Engineering)
- The physical properties considerations for design (Mechanical Engineering)
- The fluid composition and properties (Material Engineering)

5.4.1. The Outcome of the Hydraulic Simulation Studies

Pipesim and Hysys tools were used for the hydraulic exercises and these define the DN sizes of the lines. The recommended DN sizes of the two lines in this phase 1 FEED are:

24" for the 6.7km BBOU to ELEP line and

36" for the 31.5km ELEP to AKABEL line

5.4.2. The Properties

The mechanical parameters shall be guided by the API specification 5L and the calculation is based on ASME 31.8 guidelines, resulting in the initial wall thicknesses consideration of using grade X70 or alternatively X65 are:

- 11.58mm for the 24" Gas pipeline and
- 17.35mm for the 36" Gas transmission pipeline.

The parameters considered are Ultimate Tensile Strength, Yield Strength, Elasticity, Elongation, Hardness, Toughness, Creep resistance and fatigue resistance.

5.4.3. The Process Fluid Properties for Pipe Materials Selection.

The line pipes and piping materials mainly depend on the process conditions such as:

- The wet gas fluid composition
- The design temperatures

- The design pressures
- Gas fluid rates
- Water flow rates
- NGC condensate flow rates and
- Impurity mole% or partial pressure of the corrodant.

This information is inputted into the Hydrocor for the unmitigated corrosion rate and hence the unmitigated CA.

If this is not acceptable an inhibition system shall be considered.

Again, if this is not acceptable, optional materials are evaluated for the materials selection process.

At the end point of the stage a material shall be selected accordingly.

5.4.4. Other non-process Factors Considered

Other non-process factors that are considered in completing this process includes:

- Cost of materials
- Availability
- Weldability
- Other metallurgical properties and
- Manufacturability.

6. Line Pipe Materials Specification

6.1. General

This document on line pipe materials specifications shall provide the appropriate line pipe materials selection based on the following:

- Outcome of the processes of hydraulic studies - the DN's shall be obtained
- Physical parameters calculations - initial WT's shall be obtained through API 5L, ASME31.8, ASTM.
- For the transported fluid product, corrosion allowance (CA) analysis shall be obtained, using NACE, and Hydrocor tools.
- From Metallurgical point of view, the metallic properties are evaluated, reference to compositional analysis.

The resultant conclusions are as follows.

The specified steel grade is API 5L grade X70 PSL 2 – low C-Mn alloy steel, more so for the critical nature of service of the pipeline.

The line pipe Wall Thickness (WT) - is calculated by the addition of the initial WT obtained, and the mitigated CA making the wall thickness of the line pipe process to be finalised.

The company (Champion X) inhibitor code CORR1 1447A (Old code – EC1458A) was selected (See section 10.)

The material selected for the line pipes is Low Carbon-Manganese alloy steel with Nb, V and Ti in the composition.

This specification shall be in accordance with API 5L & ISO 3138 and the sub components pipe and piping classes shall be in accordance with ASTM, ASME and relevant Shell DEPs on the subject. This piping class and classifications will make it possible to avoid the unnecessary efforts during engineering fabrication, procurement shipping, transportation storing of line pipe, the requirements of ancillary requirements fittings and other equipment shall be linked with appropriate piping class.

There are two-line pipes addressed in this document:

- The 24"x6.7km BBOU to ELEP line
- The 36"x31.5km ELEP to AKABEL trunk line

6.2. Purpose

The materials specification will be used to define the guidelines, procedures, codes and standards for the line pipe, covering design, engineering, procurement, fabrication, installation and commissioning. It will link the piping class to the subcomponents of the pipelines.

6.3. 24" BBOU to ELEP line and the 36" ELEP to AKAB Line Specification

6.3.1. Information and Data for the Pipelines

Information on requirements of the selected materials from the design are tabulated in Table 5.1 below:

Table 5.1. Information and Data for the Pipelines

Description	Data	Units
Design Life Time	30	Years
Codes	ASME 31.8, ISO 3183, DEP 31.38,01.84 Gen (Materials Selection Index)	
Service/Duty	Transportation of Wet Gas	
Design Factor	0.6	
Design Temperature	45 to 50	°C
Inlet Pressure	100	barg
Pressure Class	ANSI 900# (Class 900) API 3000	
Test Pressure	1.25 x Design Pressure	psi
Maximum Allowable Operating Pressure	N/A but less than design pressure	
Spacing of Existing Line	No existing line	
Pipe Length (Survey Length),	6.7 / 31.5	km (BBOU-ELEP / ELEP-AKAB)
Pipe DN (Hydraulic Studies)	24 / 36	inch
Wall Thickness from Physical Parameters	15.88 /23.83	mm
Product Fluid	Wet Gas	
Product Impurity/ Water Analysis	No - H ₂ S, Cl, HCO ₃	mole%
Gas Flow Rate (24" / 36")	(340 / 340)	MMscfd
Water Flow Rate (24" / 36")	(9/16.5)	Mbwpd
Corrodant / Corrosion Threats	CO ₂ - Sweet Corrosion	
Corrosion Rate (Unmitigated) Hydrocor	1.85 / 3.93	mm/yr
Unmitigated CA	55.5 / 117.9	mm
Mitigating with Inhibitor CA/(Considering the ability of over 25% Corrosion resistance on selected materials)	4.2 (3.) / 8.8(6)	mm
Material Selected 24"/36"	API 5L Gr x 70 Sch. 160	Low C-Mn alloy
Manufacturing Tolerance	+10.5 to - 3.5	%
The length of Line-Pipe joint.	40 (12)	Ft (m)
Number of Line-Pipe to be manufactured (Additional lines covering post manufacturing defects and sparing are not included.	24" BBOU to ELEP line = 560 (Plus 10%= 616) 36" ELEP to AKABEL =2,625 Plus 10% = 2,888)	
Longitudinal welding	to be advised by manufacturer	
Pipe End	BE	

6.3.2. The Chemical Composition

The chemical composition of the line pipe API 5L Gr x 70 Sch. 160 is specified according to ASTM A234 S3 applied with the following restrictions in Table 5.2 below:

Table 5.2: The Chemical Composition

Specification: API 5L PSL 1/2										
Steel Grade: Grade X70 (L485)										
Chemical Composition (%)							Mechanics Capability			
							Pipe body			Weld seam
C max.	Mn max.	P max.	S max.	V max.	Nb max.	Ti max.	Yield Strength min. (Mpa)	Tensile Strength min. (Mpa)	Elongation (%)	Tensile Strength min. (Mpa)
0.26 ^e	1.65 ^e	0.030	0.030	F	f	F	485	570	c	570
a. $Cu \leq 0.50\%$; $Ni \leq 0.50\%$; $Cr \leq 0.50\%$; and $Mo \leq 0.15\%$ b. For each reduction of 0.01% below the specified max. Concentration for carbon, and increase of 0.05% above the specified max concentration for Mn is permissible, up to a max. of 1.65% for grades $\geq B$, but $\leq X52$; up to a max. of 1.75% for grades $> X52$, but $< X70$; and up to a maximum of 2.00% for X70. c. Unless otherwise agreed $Nb + V \leq 0.06\%$ d. $Nb + V + Ti \leq 0.15\%$ e. Unless otherwise agreed. f. Unless otherwise agreed, $Nb + V = Ti \leq 0.15\%$ g. No deliberate addition of B is permitted and the residual B $\leq 0.001\%$										

6.3.3. Procurement / Manufacturing Specification

This specification covers the procedures that shall be contended with in the process of manufacturing the line pipes. The following are standards and codes that are needed for the manufacturing:

Table 5.3: Manufacturing information for Requisitions

	S/N	Items	Description
General	1	Service	Wet Gas Transportation
	2	Vendor / Local Agent	VTA
	3	Manufacturer	VTA
	4	Project Name	Gas Processing and Methanol Complex
	5	Location	BUBOUWE – ELEPA - AKABEL, Bayelsa State
Design Information	6	Design Life	30years
	7	Design Factor	0.6
	8	Design Pressure (barg) Inlet	80 to 100
	9	Design Pressure at Gas Plant (barg)	60 to 65
	10	Pipeline Length (km)	6.7/31.5 (totalling 38.2 km)
	11	Gas Flow Rate (MMscfd)	340
Materials	12	Line pipe Specs (MS)	24"/36" API 5L Gr x 70 Sch 160 /
	13	Materials Standards and Codes	ASTM A53 Type F or Type E/
	14	Line pipe Length (ft/m)	40/12
Manufacturing	15	Process of Manufacturing	VTA (Basic oxygen steel process, electric arc smelting or acid open hearth process)
	16	Longitudinal welding	VTA {ERW (HFI), SWA Helical/Spiral weld}
	17	Mill Tolerance (WT)/Ovality/Length	(+10.5 to -3.5%) / $\pm 1\%$ / 12m
	18	Marking	The pipe shall be naked to ensure full traceability to melt and heat treatment lot.
	19	Certification	Quality system certify ISO 9001. Inspection witnessing
	20	Tensile testing / hardness / toughness	On cut out specimen
	21	NDT	ES
	22	Hydrostatic Test	Yes
	23	Provision of end cap	Yes

6.3.4. Fabrication

This fabrication specification shall address all additional materials to be used to enhance the protection quality, flotation control of the line pipes and their sub-components to be provided to

allow total completion of the pipeline design works prior to the construction practices. These shall include:

- Materials for external protection. (This is covered in Section 14 of this document (Coating and CP)).
- Components – fittings, elbow, bend, flanges, gaskets, etc shall be fabricated in accordance with the latest edition of the standards and pipe classes.
- Equipment that shall be connected into the line shall be implemented to provide control of galvanic corrosion. The permissible variation in outside and pipe end preparation shall be executed before they are connected into the pipeline systems.
- Floatation weight loading (concrete addition) should be incorporated as required by the design.

The specification shall address two main fabrication systems:

- Shop Fabrication – complemented by shop drawings, procedure guidelines and documentations. Relevant inspection monitoring and witnessing covering fabrication and acceptance tests.
- Site Fabrication – All site fabrication shall be carried out during or before construction work with similar approach as for shop fabrication including HSE and well-developed security system needed in Bayelsa State.

6.3.5. Handling, Transportation and Stacking of Line pipes

This specification addresses the codes and standards for transportation, storage and stacking including inspection and repairs and records of coated line pipes.

It shall always be handled in safe organized manner ensuring no damage is sustained. All pipes shall be clearly identified, labelled, and set aside for further examination. The stacking of pipes shall be done with wooden spacers in accordance with international standards and manufacturers specifications.

The following API Recommended Practices shall be followed:

API 1169 – Pipe Construction Inspection

API 5L1 – Recommended Practice for Rail Road Transportation

API RP 5LW – Recommended Practice for Transportation of line pipes in barges

7. Piping Materials Specification

7.1. General

All piping materials along with their sub-components are specified in accordance with their piping class and classification according to the following international standards and codes: NACE, DEY, ISO, EEMUA, DNV, ASTM, ASME and DEPs, PD5500, EN13480, BS4515, EN10111, API510, API620, ISO 15614-5 2004

As for other cases the links from input information from the process studies, the P&ID, identifying numbering methodology, sizing and then the initial thicknesses calculation by mechanical engineer shall be used for the required data.

Then, the corrosion/materials engineer reviews the impact of the environment internally and externally and follows the process contained in the materials selection philosophy to come up with a choice. The choice is complemented with piping classes and other classifications to reduce the unnecessary effort of going through the whole process except on special cases of materials consultancy basis, then the whole process is rounded up. For general usage these standards, - Shell DEPs: 31.38.01.12-Gen – Manual Piping Classes – Refining and Chemicals, 31.38.01.10-Gen – Manual Piping Classes – Basis of Design; ASME pressure rating, etc, covers the specification requirements.

To complement the needs for special services in piping material selection processes, the need to review piping and piping classes are effected in the FEED to ensure that technical integrity is achieved.

7.2. Piping Classes

Piping classes shall be applied in accordance with Shell DEP 31.38.01.11-Gen – Manual – Piping – General Requirements. The identification methods to be used, are given in three parts below:

Part 1: In the ASME rating classes: (1) Class 150, (3) 300, (6) 600, (9) 900, (25) 2500.

Part 2: Very significant to this standard is the number allocated to the materials selected for the project which is given as: 1 – (CS), 2 – (low and intermediate alloy steel), 3 – (SS), 4 – (Aluminum / alloys), 5 – (copper and copper alloys), 6 - (Nickel based alloys), 7 – (Non-metallic materials), 8 – (carbon steel and /or galvanised systems).

Part 3: Indicates a process related to the impact testing and non-impact testing. (ASME B31.3)

7.3. The Principles

In principle the specification shall be developed from the design output, P&ID etc and relevantly spells out the requirements for procurement, manufacturing & fabrication, construction, pre-

commissioning & commissioning, project monitoring, HSE and security. Significantly, the piping class and classification shall be used for ease of good decision making and reduction of work load. Essentially, suppliers shall be integrated into the decision-making process which will create incentive to optimize supplier's specification, manufacturing and operations to the long-term benefit of the project and in accordance with following standards:

- DEP 30.10.01.11-Gen: Manual - Metallic Materials – Selected Standards
- DEP 30.10.02.13-Gen: Manual – Non-Metallic Materials – Selection and Application
- DEP 30.10.02.31-Gen: Technical Specification – Metallic Materials – Prevention of Brittle Fracture

7.4. Manufacturing

Required information shall be prepared and requisitioned to the manufacturer with piping class / classification well defined. All drawings required, shall be approved accordingly.

7.4.1. Tolerance

The tolerance is normally 12.5% to – 3.5% but the preference should be given to +10.5 to - 3.5%. The allowable stresses should be in accordance with ASME B31.3 whilst the branch connections should be in accordance with ASME B31.1.

7.4.2. The heat treatment

PWHT should be agreed by the principal and carried out in accordance with international codes and standards, - AWS.

7.4.3. Handling, Haulage and Storage

Care shall be taken in such a manner that the handling haulage and storage shall not lead to damages of piping and their components. Preservation, protection and security are relevant if materials are to be stored for a long period of time.

7.4.4. Quality

All inspection, laboratory tests and other tests, witnessing shall be carried out in accordance with ISO 9001.

7.4.4.1. Longitudinal Welding

This shall be done according to ASME, and AWS standards and small size pipe will be seamless. Bigger sizes above 20" shall be ERW, or LSAW, or DSAW, etc and shall be in accordance with ASTM A53, 105, A106 and ISO 15614-5:2004.

7.5. Construction Processes

Construction shall be carried out in accordance with the AFC drawings, the specifications approved by the principal, along with international codes & standards. All procedures and

method statements shall follow the quality assurance processes as will be required on site.

The following are relevant to the construction practices

- Erection processes in accordance with BS 5950 – [Structural Use of Steelwork in Building](#)
- Bolt Tensioning in accordance with DEP 70.08.10.11-Gen: [Manual – Mechanical Maintenance Equipment, Tools and Bolt Tensioning](#). This also shall cover studs, nuts, washers when they are used.
- Socket-Welded Piping into Components
- Mono Flange and other flanges Processes
- Instrument Valve

7.5.1. Pre - Commissioning and Commissioning

Pre-commissioning test and check list will commence by the contractor preparing a list that would be approved by the principal specifying:

- a. Piping number and services
- b. Actual tests and checks to be carried out shall cover condition and duration of the tests
- c. Results and other data shall be obtained during the test and checks
- d. Name and function of persons conducting and witnessing the tests shall have space for signatures

For mechanical and instruments completeness, non-conformance and corrective actions shall be followed through to completion.

7.5.2. Operation Processes

Before the handover of the piping systems to operators of the process, all test completions shall be verified including pressure testing to 1.25% of the design pressure. Documents and drawings ('As-built') are part of the hand over process.

All the remedial actions that are highlighted in the technical pre-start up audit report, shall be closed out before the startup of the operation.

8. Equipment Materials Specification

8.1. General

This Phase 1 FEED identified the equipment that shall be put together for this project. Their specification documents, the methods of production capacity, power requirement, fabrications methods, and other finer details that make them appropriate for use to produce world-class facility to meet the customer's requirements at optimal economic cost.

It shall stipulate:

- Handling of process conditions
- Purchasing of equipment
- Electrical power requirement
- Specifications and standards covering all equipment and components for this project in accordance to international and industrial best practices

They also include:

- Vessels (pressure)
- Tanks (injection systems)
- Valves
- Instruments (gauges, meters)
- Electrical systems equipment
- Pumps
- Other inter-connecting components.

The equipment material specification shall provide information and services to the principal, manufacturing and fabricating companies, vendors/agents, construction company's operators. Furthermore, it will emphasise delivery processes, technical requirements, safety and quality.

Generally, the types of specifications to be handled, shall be:

- Materials specifications
- Standards codes and specification
- Interface specification
- Test specification
- Performance
- Quality specification.

The information required to develop to the specification are to include:

- Basis of Design
- Design output

- Material requirements
- Interface specification
- Testing methods
- Performances (KPI)

8.2. Generic Equipment Specifications

The generic equipment specification will define the guidelines, codes and standards, procedures, along with maintenance and operations requirements from development to maturation through to abandonment.

The description will at each stage of the system, ensure integrity of the equipment and provide the following:

- Detailed description of the equipment.
- Package of input needed to keep the equipment going through its life time (including – consumables, installation, training and after sales)
- Description according to types and class
- The quantities required for purchase
- The function and functionality for each equipment
- The capacity
- The dimensions
- The materials
- The technical use
- Technical processes in manufacturing, fabrications, transporting, quality and test certifications and acceptance. and
- Any other relevant Information

To translate all the information there is need to prepare datasheet, this datasheet shall be used for engineering, vendors, manufacturers, fabricators and suppliers RFQ etc.

The datasheet shall be effected in accordance with the following standards and codes – SAE, NIST, ISO, IEC, CEN/CENELEC.

BS 7373 – 1: 2001 Guide to preparing specification

BS 7373 – 3: 2005 Product specification.

8.3. The Information on Data Sheet

Below is a typical framework for Datasheet (Sample) that can be used for information purpose.

This information will be collated from materials selection, P&ID and other Specialist inputs (Mechanical, Electrical, Instrumentation) etc.

Table 7.1: Sample Datasheet for use for Fabrication Information

GAS PROCESSING & METHANOL COMPLEX PROJECT			
DATASHEET FOR.....			
EPIC ATLANTIC LIMITED		Doc. No.....	1 of 1
		EMPLOYER: BRASS FERTILIZER AND PETROLEUM COMPANY LIMITED	
LOCATION		PRODUCTION FAC.	
GENERAL			
	General Information		
1			
2			
3			
4			
	Fluid Data		
5			
6			
7			
8			
	Design Data		
9			
10			
11			
12			
	Mechanical Data		
13			
14			
15			
16			
	Manufacturing/Fab. Data		
17			
18			
19			
20			
21			
22			
23			
24			
	Driver		
25			
	Utility		
26			
NOTES			

8.4. Specification for Installation, Construction, Commissioning, Handover.

Specifications shall cover the following and the standard procedures:

- General Preparation: The standards and codes shall be in accordance with BS5950 – part 2 using only new materials.

In addition, documents and Approved for Construction (AFC) drawings shall follow the following processes below:

- Mounting of Plants and Equipment (standards to be used - ISO/TC 195, ISO 8091(2008), ASME – B-30.)
- Erection Processes (standards to be used are – ISO 7205 (1974), ISO 10721 (1999), API 600, API 623, BS 4604).
- Alignments and Couplings (ISO 11630 (1979), API 686 (2009).
- Bolts, Studs, Nuts, and Washers (ASTM A354, ASTM A449, ASTM A490)
- Welding Processes (DEP 61.40.20.30 Gen, DEP 30.10.60.30 Gen, API STD 1104)
- Materials for Thermal Insulation and Cladding
- Checklist and Test Sheets (for the pre-commissioning, commissioning procedures, pre – start-up safety audit covered by BS8552-2012)
- Pressure Testing (ASME B31.1, ASME B31.3,)
- Test on Lifting Equipment - Cranes and Hoists. (DPR Guide 0024 of 2020)

9. Isolation Joint Specification

9.1. Electrical Separation System Specification

There may be wide range of electrical separation system existing. For this FEED the restriction is for the cathodic protection system – ICCP /SA current leakages into structures that are not meant to be protected by CP or as a control for galvanic corrosion.

9.2. Purpose

The purpose is to establish the materials for CP isolation electrical separation systems.

9.3. Isolation Joint

The isolation joints shall be of the Monobloc type, on the manufacturers inspection and test certificate shall be sent to the BPFCL well before the unit are due to be installed.

Isolation joint shall be internally coated by the manufacturer over the full length using an amine cure epoxy coating, applied on a blast cleaned surface (ISO Sa 2¹/₂, 8501-1, NACE No. 2).

All sealing and 50mm at each end shall be left uncoated to allow welding.

The test certificate shall conform to the following for each joint unit:

The joint shall have:

- a. Passed a hydraulic test pressure 50% in excess of the specified design pressure when freely supported.
- b. Withstood a voltage of 1500 volts AC, 50 Hz, across the joint for 5 minutes with no breakdown of insulating materials.
- c. The electrical resistance across it to be more 1 Mega ohm (1,000,000 ohm)

The fabricated welds shall have satisfactorily passed visual, radiographic and ultrasonic inspection.

An M10 size terminal stud, complete with nuts and washers, shall be attached to the stud on each of the joints.

9.4. Insulation Flange

This insulation flange maybe an alternative to insulation joint. Flanges and insulating kits shall meet the design requirement of the pipeline. Insulating gasket or raised face flanges should be the neoprene coated phenolic resin type. Bolts should be insulated using full length phenolic or composite sleeves (eg tufnol etc) and phenolic resin insulating washers and steel washers at both sides.

Insulated flanged should be assembled and tested before being welded into the pipeline. A voltage of 1500 volt DC shall be applied across the flange assembly for one minute without

causing breakdown of the insulation or flash over. Subsequently the resistance across the assembled flange shall be measured and shall be more than 1 mega ohm.

On existing flanges, the bolt-to-opposite-flange resistance shall be measured (minimum bolt to flange resistance shall be 100,000 ohms.) and the overall effectiveness of the isolated flange determined after cathodic protection has been applied. Minimum bolt to flange or flange to flange resistance shall be 100,000 ohms. This can effectively be done by measuring the current through the attached pipe using a "Swain-type" current clamp/meter.

Insulated flanges shall be protected against ingress of dirt and moisture by the application of flange protectors or protective tape as defined in international standard specification.

9.5. Flange Insulation gasket to prevent dissimilar metal corrosion.

This material specification is to isolate two different metals in contact. The electrolyte insulation is achieved through the use of flange insulation gasket kit that are design for electrically insulating the flanges. They act as an insulator between dissimilar metallic flanges, in this case DSS from SPDC Manifold Platform and MS from BPFCL Manifold Platform.

Again, CP current is isolated from casing using isolating materials.

10. Thermal Insulation Specification

Thermal Insulation is not applicable, and no project information to use here.

11. CP Design And Specification

11.1. General

This CP Design & Specification is for the ICCP and temporary design Sacrificial Anode installation for the full protection of the subject lines:

- 24" BBOU – ELEP Gas line and
- 36" ELEP – AKABEL Gas Transmission Line

The inclusion of the temporary protection is justified by the need of stringent corrosion management of the MS lines that may be expected to lose as much as 1.5mm during the construction period of 2 years. This loss is undesirable and must be prevented and not allowed to occur. This protection is essential before the energising of the permanent ICCP system because of the extremely high corrosion rate indicated from the resistivity survey of the soil environment of Bayelsa state. The table 10.1 gives some needed information for the design.

11.2. Design parameters

a. Pipeline Information

Table 10.1: Pipeline information

Pipeline Technical Information Input / Data	BOBOUWE to ELEPA Line	ELEPA to AKABEL Line
Pipeline Grade	X70	X70
DN	24" (0.61m)	36" (0.91m)
Coating Resistance	18 ohms	18 ohms
Design life (Temporary CP/Permanent CP)	3yrs/30yrs	3yrs/30yrs
Pipeline length	6.7km (6,700m)	31.5km (31,500m)
Pipe coating efficiency (uncovered Area % coating breakdown)	Initial-2, Med life-5, Final life-7	Initial-2, Med life-5, Final life-7
Average current density	22 mA/m ²	22 mA/m ²
Pipe joint Length	12 m	12 m
Design Pressure	110 barg	110 barg
Temp.	45-50 °C	45-50 °C
Ambient Temp.	35 °C.	35 °C.
Wall Thickness	15.88 mm	23.83 mm
Selected Coating	3LPP	3LPP

b. Expected Coating Breakdown

Polypropylene coating breakdown value is 5 to 7% at 25 years, and thereafter will be at an expected 0.75 to 1.5% per year, judging from DnV B-401 standards.

- c. Reference Electrode Potential Ranges for Steel Protection (starting with the criteria for Protection to the beginning of overprotection) is given in the table below.

Table 10.2: Criteria for Protection Vs Reference Electrode

Environment	Reference Electrodes		
	Cu/CuSO ₄ (mV)	Ag/AgCl (mV)	Zinc (mV)
Free Seawater Environment	-850 to -1000	-800 to -1050	+250 to +0
Anaerobic Environment	-950 to -1100	-900 to -1050	+150 to +0
High strength Steel UTS>700N/mm ²	-850 to -1000	-800 to -900	+250 to +100
Land/ Swamp Environment	-850 to -1150	Not used for land	Not used for land

11.3. Current Requirement

11.3.1. Current Density

Current density is the corrosion current released per square area of a material under electro-chemical influence. It follows that the total corrosion current will be the current density multiplied by the entire surface area of each of the specified lines. This area can be significantly reduced by application of coat. Allowing only for exposed areas that the coat has not been affected by corrosion. Hence the need for protection to stop corrosion using CP system. The Cathodic Protection being the application of the reverse cathode current to reduce or arrest the electro chemical corrosion reactions.

Of essence, is the determination of the total current to be used to evaluate the design properties:

$$I_c = JF2\pi rL \text{ —————eq. 1}$$

Where J – is the design electrical current density

F – Coating Breakdown Factor

2 π rL – Surface Area

11.4. Current Demand Calculations

From above equation 1, the design electric current demand for each steel bulkline & pipeline taking into cognisance the coating breakdown factor is shown in table below. They are evaluated to be 0.2 mA/m² in accordance with DEP 30.10.73.31-Gen: [Technical Specification – Design of Cathodic Protection Systems for Onshore buried Pipelines](#). Again, it should be noted that three relevant breakdowns are used for design. These included,

- The Initial Breakdown

- The Mean Breakdown
- The Final Breakdown

The current need for group of pipelines which will be hooked at the manifold from same field or area is given by equation 2

$$I_{cmf} = \sum_1^n I_c \text{ —————eq. 2}$$

Table10.3: Surface Areas and Current Demand for the Pipelines CP.

SURFACE AREA & CURRENT DEMAND CALCULATIONS										
SN	P/LINE	LINE DIA.	LENGTH (m)	BARE SURFACE AREA (m ²)	MEAN EXPOSED SURFACE AREA (2%)	EXPOSED SURFACE AREA (LIFESPAN) (5%)	CORR. CURR. DENSITY (mA/m ² Bare Steel)	FINAL CURR. DEMAND (Amp) 7% used for Cal	MEAN CURR (Amp)	INITIAL CURR. (Amp)
1	BBOU – ELEP	24" (0.61m)	6,700	12,833.18	256.66	641.66	22	19.80	14.12	5.6
2	ELEP – AKAB	36" (0.91m)	31,500	90,008.10	1,800.16	4,500.41	22	138.6	99.0	39.6

11.5. Sizing of Transformer – Rectifiers

11.5.1. Transformer – Rectifier for group of pipelines (n)

The transformer rectifier is used to convert AC to DC which is required for CP.

The size is based on the current demand and addition 50% and then using ohm's law, the voltage is obtained reference equations 3 & 4

$$I_{CTR} = \sum_1^n I_c + 50\% \sum_1^n I_c \text{ —————eq.3}$$

The Ohm's law which is given as

$$V = IR \text{ —————eq.4}$$

Where, V is Voltage, and R is Resistance. The resistance is based on the entire CP system resistance

Table 10.4 Design for Temporal Anodes Materials

Anode Material	Mg alloy
Current Density	22 mA/m ²
Dimension	146m x 472m
Mg alloy consumption rate	4kg /amp-yr

Potential with ref to CuSO ₄	-1.55 volts
Electrochemical capacity of Anode	1,230 Amp-hr/kg
Weight of Anode/Packed weight	14.5kg / 30kg
Utilization	0.85

Equations for calculation are as follows:

a. From BBOU to ELEM pipelines

$$\text{Current (amp)} = \text{Protected Area (m}^2\text{)} \times \text{Current density (mA/m}^2\text{)} / 1000 \dots\dots\dots \text{eqn. 5}$$

$$\text{Weight (kg)} = \text{Current (Amp)} \times \text{design life(years)} \times 8766 / \text{Capacity of material (Amp/kg)} \times 0.85 \dots\dots\dots \text{eqn. 6}$$

$$\text{Number of Anodes} = \text{Weight Required} / \text{individual Net Anode Weight} \dots\dots\dots \text{eqn. 7}$$

$$\text{Number of Anodes} = \text{Current Required} / \text{Individual Current Output} \dots\dots\dots \text{eqn. 8}$$

$$MA = It \times td \text{ Ca/u} \dots\dots\dots \text{eqn. 9}$$

Where Ma = mass of Anode = 14.5kg

Td = Design life (year)

It = Current demand (Ampere)

Ca = Consumption Rate of Anode (kg/A – y)

U = Utilization factor

$$MA = It \times td \text{ Ca/u}$$

$$MA = 5.6 \times 3 \times 4 / 0.85$$

$$MA = 79.10$$

$$\text{Number of Anodes} = 79.10 / 14.5 = 5.45$$

Number of Anodes is approximately 6 for 24" BBOU to ELEM pipelines.

b. From ELEM to AKABEL pipelines

$$\text{Current (amp)} = \text{Protected Area (m}^2\text{)} \times \text{Current density (mA/m}^2\text{)} / 1000 \dots\dots\dots \text{eqn. 5}$$

$$\text{Weight (kg)} = \text{Current (Amp)} \times \text{design life(years)} \times 8766 / \text{Capacity of material(Amp/kg)} \times 0.85 \dots\dots\dots \text{eqn. 6}$$

$$\text{Number of Anodes} = \text{Weight Required} / \text{individual Net Anode Weight} \dots\dots\dots \text{eqn. 7}$$

$$\text{Number of Anodes} = \text{Current Required} / \text{Individual Current Output} \dots\dots\dots \text{eqn. 8}$$

$$MA = It \times td \text{ Ca/u} \dots\dots\dots \text{eqn. 9}$$

Where M_a = mass of Anode = 14.5kg

T_d = Design life (year)

I_t = Current demand (Ampere)

C_a = Consumption Rate of Anode (kg/A – y)

U = Utilization factor

$M_A = I_t \times t_d \times C_a / u$

$M_A = 39.6 \times 3 \times 4 / 0.85$

$M_A = 559.10$

Number of Anodes = $559.10 / 14.5 = 38.60$

Number of Anodes is approximately 39 for 36.5" BBOU to ELEP pipelines.

11.6. The sizing of Transformer Rectifiers

The formula controlling factor is just the current as all the will be relatively low the controlling input will be the current. And the other factor the system in the Resistances of the system. At the low resistivity obtained from the survey. It will be noted that it will be less than 0.5 ohm. ie the limiting factor will be the current. The limiting current of a TR is 100Amp. and the voltage is 50V. Hence, using the eqn..3 the numbers of T/R needed are:

- i. One with a limiting current of 29.7Amps and less than 15V at BBOU Manifold at the final lifespan of the project. At the start of the project this is much less. So TR select will be 30amp/15V or the next bigger commercial unit, i.e. 50A/25V
- ii. Following the same approach the current needed for Elep - Akabel line is 208A at the end of the project ie Two TRs will be needed with the limits of 100A /50V. For better spread of current one will be placed at Elepa and the other at Akabel.

11.7. Groundbed Design

The design was based on the rule of thumb and the local vendor / manufacturer's advice and instructions on the use of the mixed anode GB system.

- At Bubouwe a six mixed metal oxide type anode is selected for design whilst
- At Elepa and Akabel a ten mixed metal oxide type anode for each site is selected.

11.8. Design Output

The table below reflects the outcome of the CP design.

Table 10.5: CP Design Output

S/N	Description	Design Output	
		24" BBOU	36" AKABEL
1	Sacrificial Anode Design	6 nos. Cir. Parked Mg Anodes (30kg)	39 nos. Cir. Parked mg Anodes (30kg)
2	Anode spacing	3M	3M
3	At river crossing beyond 2km	Al bracelet Anodes	Al bracelet Anodes
4	Test station	1 no. at 1km interval	1 no. at 1km interval
5	Transformer Rectifier	1 no. 50amps/30v TR	2 nos. 100amps/50v TR (one at ELEP and the other at AKABEL)
6	Junction Box/5ways klipon Box	1 nos.	2 nos.
7	Groundbed	1 no. of 6 mixed metal oxide Anode	2 nos. of 10 mixed metal oxide Anode
8	Cables	Refer to drawings	Refer to drawings
9	Insulation		

12. Corrosion Inhibitor Specification

12.1. General

The impurities compositions of the wet gas from reservoirs are given in table 11.1 below. The fluid products indicate susceptibility of CO₂ or sweet corrosion. This corrosion threat to MS, was rated for corrosion using initially the De – Waard- Milliams formula and further compared with extrapolation using Nomogram (Ref. fig 11.1) and they were found to be high.

Table 11.1: Reservoirs / Flowing Conditions & Impurities Compositions (Water Saturated Compositions)

Component	Impurities Compositions - Mole %				
	BBOU	ELEP	NUNR	SIEB	OPUG
N ₂	0.07	0.00	0.00	0.00	0.07
CO ₂	0.59	2.18	0.77	0.77	0.59
H ₂ S	0.00	0.00	0.00	0.00	0.00
H ₂ O	0.34	2.19	0.63	0.95	1.08
Total	1.00	4.36	1.40	2.48	4.11
Reservoir / Flowing Conditions					
Res Pres (barg)	223.8	476.6	500.6	450.6	356.9
Res Temp (°C)	78	123	102	117	113.8
FTHP (barg)	100 - 80	100 - 80	100 - 80	100 - 80	100 - 80
FTHT (°C)	62 - 58	63 - 60	63 - 60	63 - 60	62 - 58
pCO ₂ (barg)	0.59 - 0.47	2.18 - 1.74	0.77 - 0.62	0.77 - 0.62	0.59 - 0.47

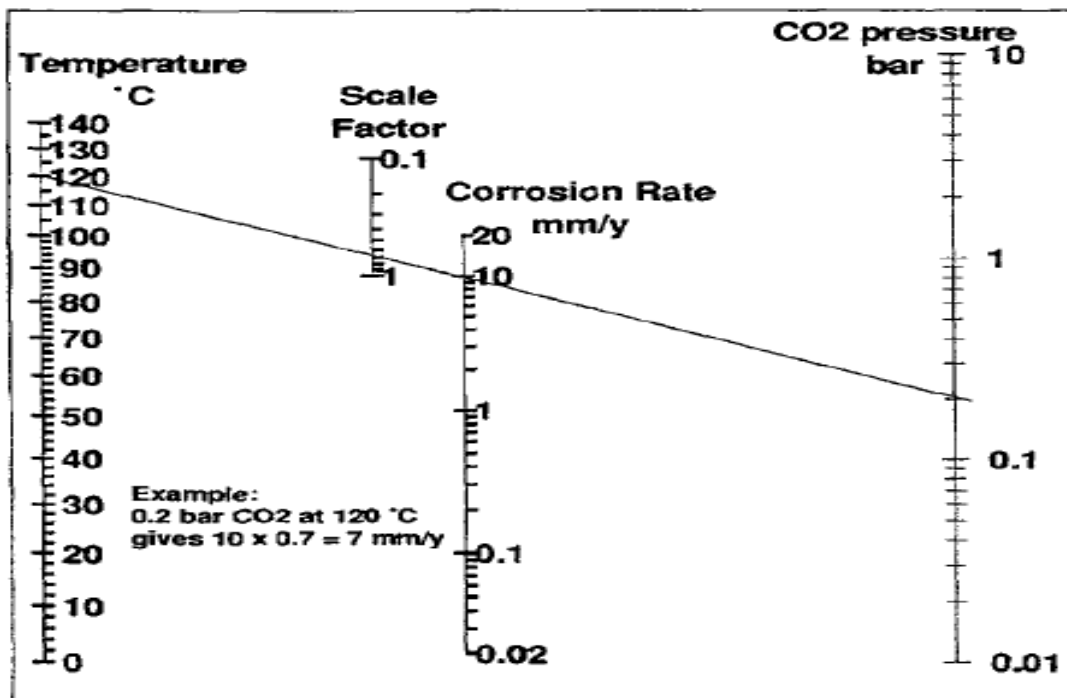


Fig 11.1: DeWaard-Milliams CO₂ Corrosion Nomogram

Again, Hydrocor software assessments confirmed high rates of corrosion (Fig 11.2 and Fig 11.3) in bare MS pipeline carried out for this phase1 FEED. The unmitigated corrosion rates are 1.85 mm/year for the 24" BBOU – ELEM Line and 3.93 mm/year for the 36" ELEM – AKAB Line. These rates shall result in CA of 55.5mm and 117.5mm unmitigated corrosion allowances for the design life of 30 years, respectively. The rates are undesirable and unacceptable. Hence, corrosion mitigation using inhibition was assessed in accordance with the philosophy in chapter 4 of this document.

The resultant outcome along with the materials' resistance to CO₂ corrosion of 25% of the selected pipeline materials (API specification 5L grade x70 low C-Mn alloy steel) is 3mm and 6mm mitigated CA for the 24" BBOU – ELEM Line and ELEM – AKABEL Line respectively. The reduction in the corrosion levels were quite high, thus justifying the use of the inhibitor. Thus, the use of inhibitor allows us to accommodate the use of the API Grade 5L X70 specification for the pipeline choice.

Consequently, the targeted reduction in CA can be achieved using the recommended Champion X's inhibitor {CORR11447A (EC1458A)} – Champion X is a reputable American chemical company with representation in Nigeria). They selected an appropriate inhibitor that will achieve a target of 92.5% corrosion reduction based on Proforma requisition from Epic Atlantic, following NACE SP 21469 Selection & Management Process for Corrosion Inhibitors.

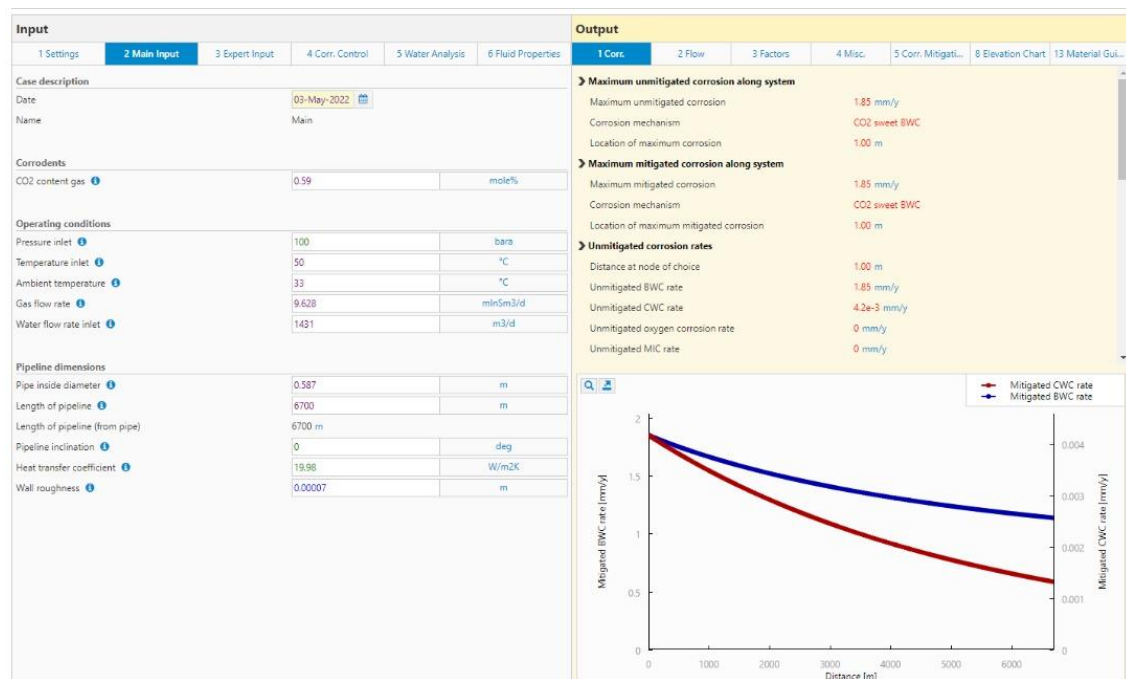


Fig 11.2 Hydrocor Report for 24" BBOU to ELEM Gas Pipeline

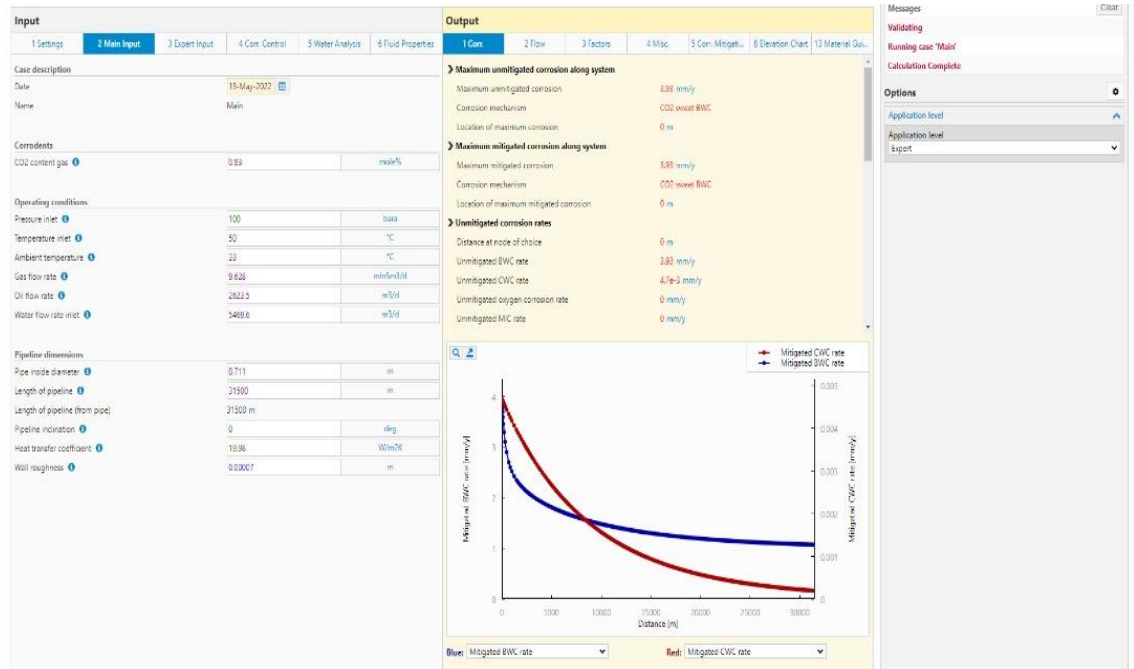


Fig 11.3 Hydrocor Report for 36" ELEP to AKABEL Gas Transmission Line

12.2. Purpose

The purpose is to select an inhibitor to achieve the goal of reducing the corrosion rate by 92.5%

12.3. Specification

Recommended inhibitor – CORR11447A (EC1458A)

Rate of injection – see table – 41 gal/day for 24" BBOU – 70gal /day for 36" ELEP- AKABEL line

Injection Discharge Practice - Continuous injection

Table 11.2: from Champion X Inhibitor Injection Rates

Corrosion Inhibitor Chemical EC1458A (aka CORR11447A)											
S/N	P/L system	Gas Prod. (MMscfd)	Press. (bar)	Temp. (°C)	Condensate (Mbcpd)	H ₂ O Prod. (Mbwpd)	CO ₂ (mole%)	Corr. Rate (mm/yr)	Diameter (in)	Injection PPM	Injection Rate (gals/day)
1	BBOU to ELEPA	331	100	45-50	16.2	9	0.59	1.85	24	109	41
2	ELEPA to AKABEL	386	100	45-50	34.4	16.5	0.83	3.93	36	101	70

12.4. Codes and Standards

NACE- 5134-3980 Standard Laboratory Tests and evaluation. This shall also address the residual output of the inhibitor resulting from the injection - DEP 31.01.10.10-Gen: Manual – Chemical Injection Facilities.

12.5. Injection Provision

A chemical Injection system shall be provided in accordance NACE and other international standards and codes.

The injection system should be designed, constructed, and operated in accordance with NACE standards and codes

The pump shall be rated for this input. The manufacturer's information is given in the Datasheet attached of this document.

12.6. Operation Practice

The recommended practice shall be for continuous injection of the inhibitor at the recommended rates.

The inhibitor shall be stored in the tanks provided at the manifold for period agreed and approved by the principal. The storage capacity shall be based on the number days the principal approved to handle the daily rates and the frequencies of replacement of the inhibitor.

Inhibition residual checks and test shall be effected on random sample at the remote end from the pumping locations at Bubouwe and Elepa.

Quality check of product must be certified and guaranteed.

The running of the system shall be done by the operation staff with support from the manufacturer.

Contamination of the inhibitor must be totally prevented and practice for storing the inhibitor shall be done according to approved international standards and codes. SHOC (MSDS) standard for the handling of chemicals shall be strictly followed.

13. CP Type Selection

13.1. General

The soil resistivity survey data carried out by the Brone Positioning and Survey Limited, showed that most of the area have very low resistivity reflecting extreme corrosion rates that could be 0.385/0.500 mm/yr. Hence, there is need to arrest the corrosion to the external surface of the pipeline at all times including during construction. Consequently, should there be no CP during the construction period of say three years, there is the possibility that the line will corrode up to 1.5mm and this is undesirable. Conclusively, there is need for temporary CP on the pipelines during the construction period.

Cathodic protection is a technique used to reduce or arrest the corrosion of a metal surface by passing sufficient cathodic current to it to cause its anode dissolution rate to stop or reduce using direct electric current (DC) from one external source to oppose the discharge of corrosion current from anodic area on a metallic structure. CP can be achieved through two main means:

- Sacrificial Anode (S.A) Cathodic Protection System
- Impressed Current Cathodic Protection (I.C.C.P) System.

Each of these systems will require other sub level selections of CP items and equipment.

Overall, the intension of this write up is to build up the CP type from the choices made from items and equipment to allow for proper CP design.

13.2. Purpose

The CP type selection is to describe the CP system, the materials and equipment that constitute the field design make up.

13.3. Criteria for Cathodic Protection and Overprotection Potentials for Pipelines.

Before we undertake a look into the two cathodic protection systems, it may be necessary to have a brief review on electrochemical measurements.

Theoretically, the thermodynamics of CP is based on electrochemical laws. These laws emphasize that potential measurement between the structure and the electrolyte environment (Soil, Water etc) are executed using half-cell reference electrodes. The table 1 below shows the range criteria for protection and overprotection level using the main commercial reference electrodes.

Table 12.1 Criteria for Protection and Overprotection Potentials.

S/N	Environment	Reference Electrode		
		Cu/CuSO ₄ (mV)	Ag/AgCl(mV)	Zinc(mV)
1	Soil environment	-950 to -1100	Not used for Soil	Not used for soil
2	Free sea water	-850 to -1000	-800 to -1050	+250 to +0
3	Anaerobic environments	-950 to -1100	-900 to -1050	+150 to +0
4	High strength steels UTS > 700 N/mm ²	-850 to -1000	-800 to -900	+250 to +100

The beginning of the range establishes the minimum potential for the CS structural protection while at the other end is the beginning of the disbanding of the coatings and paints. Consequently, at this range, protection without resultant damage to the coat.

13.4. Sacrificial Anode Type Cathodic Protection

From the indication of the corrosion rate, 1.5mm deterioration can occur at the construction period of three years and a temporary protection of sacrificial anode shall be used to arrest the deterioration of the pipelines externally.

13.5. The Principles of Sacrificial Anode (S.A) Cathodic Protection.

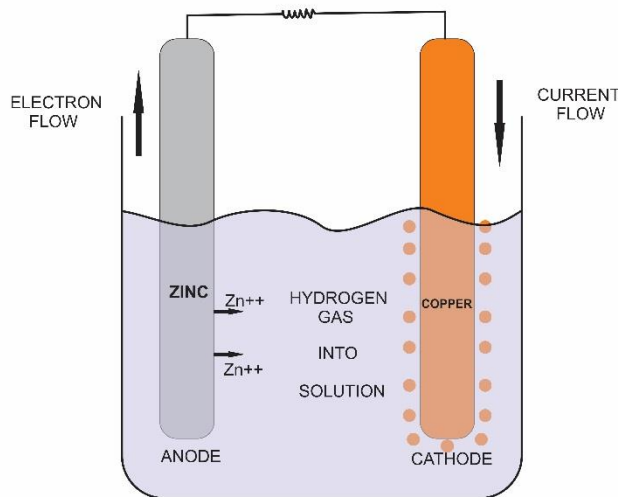


Fig 12. 1 - When zinc and copper are electrically coupled and immersed in a hydrochloric acid solution

The basic idea behind sacrificial anode protection is the selection of two dissimilar metals from the galvanic or electrochemical series (ECS) with the metal higher in potentials in an electrolyte sacrificing itself to protecting the metal in the lower ECS.

13.6. Selection of Sacrificial Anode Materials:

There are three main commercial anodes that exist from which this selection of S.A. can be

made, depending on their peculiarities and the environment:

- Magnesium Anode
- Aluminium Anode
- Zinc Anode

Table 12.2 Selection of Sacrificial Anode at Different Environment

S/N	ANODE	ENVIRONMENT						
		SOIL RESISTIVITY			WATER			
		Low	Moderate	High	Salt	Brackish	Fresh	Remarks
1	Mg	=	✓	✓	X	X	✓	Most suitable for Soil
2	Al	X	X	X	✓	✓	✓	Most suitable for low resistive water
3	Zn	=	=	X	✓	=	X	Mainly used in sea water

Legend – ✓Fit-for-purpose, X-Not Acceptable, =may be used

Table 12.3 Comparison of Anodes

S/N	Anode Type	Same Weight Ratio of Consumption	Relative Density	Cost Of Anode Same Weight (Ratio)	Installation Cost
1	Magnesium	3	27	5	1
2	Aluminium	13	42	3	1
3	Zinc	10	100	8	1

Conclusively, the selection of sacrificial anode for this C.P. type for the soil resistivity in Brass area is Magnesium anode; and for large river crossing aluminum bracelet anode.

13.7. Impressed Current Cathodic Protection

Impressed current is a type of cathodic protection utilizing electrochemical means to obtain protection against corrosion. This reduces the dissolution of anodic structures by decreasing the difference between potential energy and the cathodic and anodic sites on the metal surface when put into a conductive electrolyte such as water, soil, etc.

13.8. Principles of Impressed Current Cathodic Protection (I.C.C.P.)

I.C.C.P. is obtained through impressing direct current into the pipeline from electric power source. The source that supply A/C current must be converted to D.C. through a transformer rectifier (T.R.) see Fig 12.2 below:

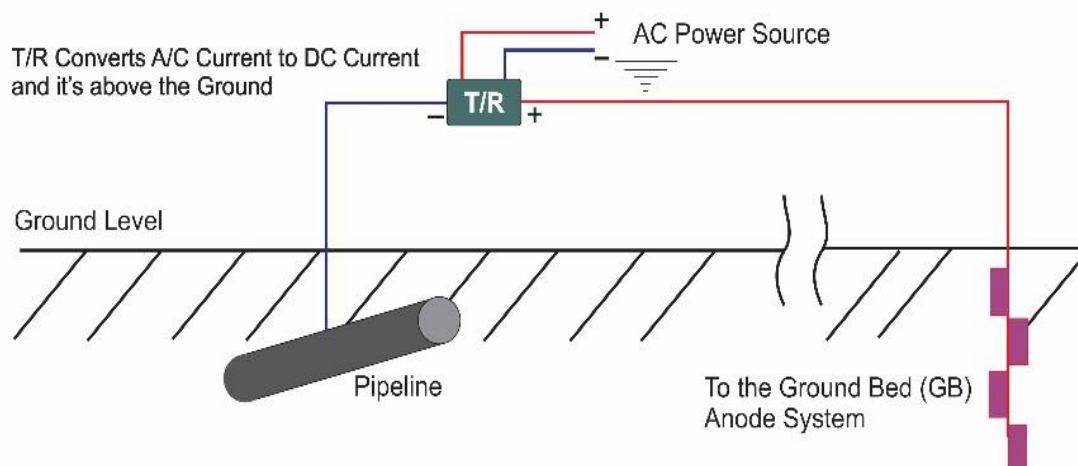


Fig 12.2. Impressed Current Cathodic Protection

13.9. Selection of Equipment Materials for I.C.C.P.

The FEED selection of ICCP type shall be based on choices made under the following sub titles:

- Power System Selection
- Groundbed Type Selection
- Groundbed location Selection
- Groundbed Anode Selection

13.10. Power System selection

The two main sources of power are namely:

- a. Alternate Current (A/C)
- b. Direct Current (DC)

13.10.1. Alternating Current (A/C)

The A/C are got from the following generating sources:

- PHED The Nigerian National Grid
- Turbo Generator (A/C)
- Engine Generator (A/C)
- AKABEL Plant (A/C)

It should be noted that only DC are necessary for the protection of structures. Hence, there is need to convert the A/C to DC for CP. Under this circumstance, the conversion is achieved by the use of transformer-rectifier.

13.10.2. The Direct Current (DC)

The DC can be gotten from the following sources:

- Solar powered system (DC)
- Wind powered system (DC)
- Thermoelectric power system (DC)
- Fuel cell (DC)
- Batteries (DC)

Cost wise, the solar panel is now getting cheaper to use at remote areas where there is no source of electricity, and therefore be considered.

13.11. Groundbed Type Selection

Basically, the types of GB systems are given below:

- a. The Typical Vertical Anodes, which will incorporate spacing along the ground as designed at equal level and intervals closed to the surface of the ground where they are buried,
- b. Typical Horizontal Anodes, - incorporating equal spacing at the burial points
- c. **Typical Deep Anode GB Type**
- d. Water Access Channel Type

In general, the deep well GB type is preferred because of space limitations.

13.12. Groundbed Location Selection

In selecting ground bed sites, the most important consideration from a design standpoint is determination of effective soil resistivity location. Below are considerations that must be considered when selecting a site.

- Are other underground metallic structures within the area of influence surrounding the ground bed? If so, they may pick up current from the ground bed and create a stray current interference problem that will require corrective measures if the site is used. This is important when planning impressed current ground beds.
- Is the proposed site on or off the pipeline right-of-way? If off (as is apt to be the case with impressed current ground beds), can a right-of-way be procured for it? Check with the principal on this.
- If a rectifier-powered impressed current system is to be installed, is there a power line present? If not, is a power line extension from the nearest source practicable?
- Is the site reasonably accessible for construction and maintenance purposes?
- Are there plans for building construction (new pipelines, highway development, or other similar work) that will make the site untenable in the near future?

13.13. GB Anode

The following are anodes that are utilized for GB systems:

- Graphite
- High silicon cast iron
- Mixed metal oxide
- Platinum
- Steel
- Lead-magnetite anode

By the rule of thumb, the most popular type of GB anode that is used in Nigeria is the mixed metal oxide (MMO) type for deep well anode system.

In addition, the selection of the GB system will incorporate cable types from the electrical source to the TR, from the TR to the GB. Again, from the TR to the pipelines at the Manifold.

13.14. Conclusion

For the permanent ICCP system the following are envisaged to be designed for:

- TR
- Deep well GB with mixed metal oxide anodes located beyond 100m away from the manifold or any metal structures.

14. CP Type Specification

14.1. General

This FEED document shall provide the specifications for the sacrificial anode practices and permanent impressed current cathodic installation practices for the BFPCL project for the phase 1 pipeline construction for the 24" BBOU – ELEP gas pipeline and 36" – ELEP to AKAB Gas transmission line. All practices undertaken shall be based on international and industrial standards.

14.2. Purpose

The purpose of this C.P type/specification is to provide document, drawing and information practices for the cathodic protection systems.

14.3. Temporary Anode Installation Practice

14.3.1. Pipeline Data

Table 13.1: Pipeline Data

S/N	PIPELINE	NPS (inch)	LENGTH (m)	AMBIENT TEMP. (°C)	WALL THICKNESS CALCULATED + CA
1.	BBOU - ELEP	24	6,700	35	12.88 + 3 = 15.88
2.	ELEP - AKAB	36	31,500	35	17.83 + 6 = 23.83

14.3.2. S.A Material Prequalification Requirements.

The Mg anode material being considered shall be pre-qualified to ensure their performance (resistivity potential as electro chemical capacity) at the operating temperature and soil conditions. Certified test reports from the proposed anode foundry or a certifying agency approved by the principal, or for previously cast anodes of identical chemistry may be submitted in lieu of pre-qualification testing specifically for this project approval of the company in advance of casting the sacrificial anodes.

14.3.3. Quality Control:

The provision of any yard fabricator inspector for all casting and fabrication operations. Or if bought off-shelf the certification shall be provided.

The procedures, specifications shall be provided by the principal/contractor for the inspection duties. The foundry type shall be stated in the quality control procedure what QC procedure it intends to or have been used and adjudged acceptability.

14.3.4. Anode Installation:

The Anode shall be positioned on to the test station along the pipeline on pre-determined spacing. The cable connection shall be installed as shown in [EA-BFPCL-GPMC-GGPL-FEED-MCM-DWG-001](#).

The anode connection and attachments shall be approved by the principal before construction is embarked.

After the installation potential survey shall be carried out. The anode connection shall be such that they can easily be connected and disconnected during and after commissioning of the permanent system.

14.3.5. Electrical Separation (IJ/IF)

The pipeline installation shall be electrically isolated from piping, foreign structure/pipelines as well as electrical and instrument earthing system.

The isolating joints/flanges shall be installed

- At both extremities of the buried section of the line/s
- Branch line if they are not part of the main line

This is implemented to stop leakage of C.P current to other and adjacent installations.

The installation shall be done under the supervision of professional C.P specialist and the guideline for installation commissioning are given in DEP 30.10.73.33-Gen and NACE

14.3.6. Anode Commissioning

The anode commissioning shall be in accordance with ISO 15589 – 2(1) section 12.

14.3.7. Anode Specification

Should be carried out according to ISO 15589-2(1).

14.3.8. Acceptance Criteria

The acceptance is as follows:

This shall be based on agreed procedures;

The contractor shall provide documents and records required by this specification and procedures.

14.4. Impressed Current Cathodic Protection Type Specification.

This specification and procedures shall satisfy the need for the FEED and installations of the following equipment

- Groundbed

- Transformer – Rectifiers
- Cable Makers
- Junction Boxes
- Test Post

14.4.1. Groundbed -Deepwell

14.4.1.1.General

A good performance of the CP system will be obtained by deep well ground bed type.

This will allow for satisfactory, spread of protective current to the pipelines. The ground bed will comprise Mixed Metal Oxide – (MMO) based anodes Type.

Schematic Sketch below (fig 13.1)

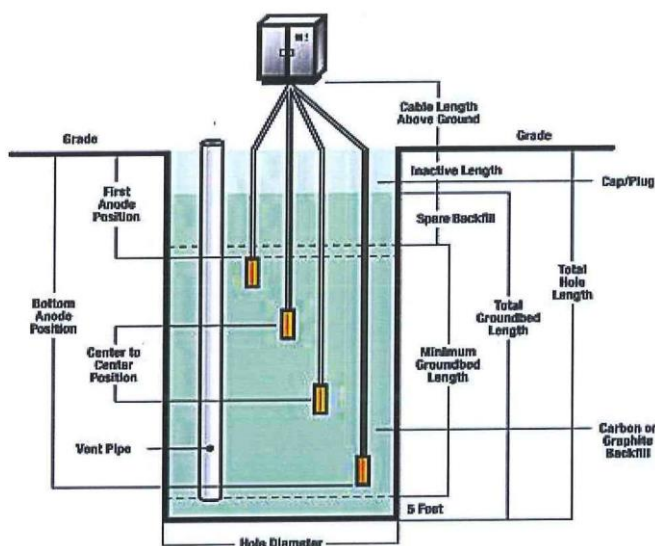


Fig 13.1: Schematic Sketch of Deepwell Groundbed

The Groundbed shall be designed according to the DEP – 30.10.73.31-Gen and NACE.

NB: The design life of the CP is 30 yrs.

The CP contractor shall decide the size of the borehole.

Each anode shall be mounted on a single self – supported single core 6 mm² EPR/PV Cable at the manufacturer's workshop.

The cable will be of a type suitable for installations in a borehole filled with carbonaceous backfill. Each anode cable shall be terminated separately at the anode bus bar.

14.4.1.2.Qualification of GB Locations

The following factors shall be considered in choosing the particular positions and type of the

groundbed:

- Soil Resistivity
- Security of the locations/vandalism or other damages
- Accessibility of the location
- Soil Moisture
- Interference to other Structures
- Proximity to source of power
- Space Factor
- Purpose of design

Hence, the GB shall be consistent with the practice below:

- CP GB shall be located at a minimum distance of 100 m from buried / submerged metal structures to be protected.
- The CP deepwell groundbed anodes shall be installed in a permanent moist wet zone / environment.
- The CP deepwell GB shall be located in relatively low resistivity soil / zone

14.4.1.3.Deepwell Materials

The materials used for vertical borehole type GB shall be:

- PVC pipe in nominal 6m length for non-active zone
- Steel pipes for boreholes “actives” lengths of casing with hack saw blade cuts or fine perforations of about 3mm holes.
- Flexible plastic (drilled) vent pipes (2” ID)
- Anode centralizers; PVC passes freely through the internal bore of PVC pipe and steel pipe.
- Concrete and steel reinforcing for borehole headwork

14.4.1.4.Installation of Casing

The size of casing shall be selected based on groundbed design.

14.4.1.5.GB Anode Installation

Anode on their cable shall be lowered into bore to the depth of the steel casing. Particular care shall be taken to ensure that the anode string is centered in the borehole. Once the anodes are inserted, centralizers are fitted as required along the anode string.

14.4.1.6.Vent Pipe

A 2” PVC pipe with ¼” holes drilled throughout the length will be provided to prevent gas blockage at the anodes.

The holes are made to correspond with the position of the anodes.

The pipe which is fixed through the anode support plate is slotted over the complete length without the loss of pipe strength. The vent pipe shall be extended through the wall of the pit and mounted in a 'U' bolt above ground grade level at the side of the pit.

14.4.1.7.Back Fills

The carbonaceous backfill shall:

- be calcined petroleum coke breeze
- have carbon content of over 98%
- have minimum bulk density of 1.04 kg/m³
- be of small diameter spherical shape with approximate 98% passing a 20 mesh screen and more than 80% retained on a 100 mesh screen.
- contain volatile contents less than 0.3%
- have a minimum specific density of 1.90

The backfill is used to surround the anodes in the active column section. It has a low resistivity to improve current distribution in the backfill column and a high content control to increase the effective useful anode life by reducing anode resistance to earth.

The steel casing will then be completely filled with low resistant carbonaceous backfill surrounding the active length. The carbonaceous backfill shall be mixed with water to make slurry free from particles and at a suitable consistency.

Coke breeze slurry shall be slowly pumped / poured from the bottom of the casing until the desired level has been reached.

The coke breeze shall form a solid column around the anodes from the bottom during the in to top. The contractor shall ensure the blockages do not occur in the anode column during the installation of the coke breeze by continuous pumping.

14.4.1.8.Borehole Headworks

Security pits shall be constructed installed at each well head (The pit measuring 1.5m x 1.5m x 1m high from the ground level will be constructed using grade 25 concrete mixed with a minimum wall thickness of (15cm). It shall be lockable with the use of high security pad lock. And protected in metal frame welded into the door

NOTE:

The pit will be constructed such that it will allow water drain off into the soil strata.

14.4.1.9.Fencing

Each ground bed site shall be enclosed with fencing in according with fence details and the dimensions consistent with the provision of well head security design.

14.4.1.10.GB Cable Marker

Cables shall be laid from the groundbed wellhead to the transformer rectifier. The route shall be traced with cable makers. They are prepositioned at intervals along the cable routes as indicated in document [EA-BFPCL-GPMC-GGPL-FEED-MCM-DWG-009](#).

The cable maker built as indicated in document [EA-BFPCL-GPMC-GGPL-FEED-MCM-DWG-002](#).

14.4.1.11.Transformer Rectifier

The TR shall be installed in accordance with [EA-BFPCL-GPMC-GGPL-FEED-MCM-DWG-009](#) at the manifold.

15. CP Materials and Equipment Specification

All the materials, equipment and their manufacturing, fabrication and other information is given in the datasheets in Attachment 1 of this document.

16. Coating/Painting Mat Specs-Linepipes/PL/Field Jnts, Structures & Equipment

16.1. General

This document is the coating/painting materials specifications for the Bubouwe manifold to Akabel location in Bayelsa state. It stems down from the conceptual design. It is the phase 1 pipelines' FEED development implementation for Brass Fertilizer Petrochemical and Company Limited (BFPCL).

It outlines the coating materials specifications which are environmentally zoned into the following areas:

- a. Buried and Submerged Lines
 - The buried line would require to be coated as part of the protection of the pipelines which is complimented by the cathodic protection systems.
- b. Splash Zone Areas
 - This zone is an area where corrosion could be very high but it does not always have the required electrolyte in continued bases. The zone is slightly above HHW and Slightly below LLW tidal zone. It needs special treatment.
- c. Atmospheric Areas
 - The atmospheric protection is normally done with marine paints for equipment above the splash zone.

In General, physical properties to be considered in selecting coatings/paints for a given application include dry film thickness, adhesion and, resistance to high or/and low temperature, and permeability to liquids and gases. The required chemical resistance obviously depends upon the environment. Since coatings vary in properties and in cost, a practical choice usually represents a compromise.

16.2. Purpose

To offer protection to the equipment, the pipelines in the established external zones.

16.3. Strategy Statement

The strategy statement is:

"The plan is to establish the best coating and paint selection in the established zones at the best economic consideration."

16.4. Scope

- Buried and Submerged Lines and Equipment
- Facilities and Utilities at the Splash Zone Areas

- Equipment and Materials at the Atmospheric Areas

16.5. Buried and Submerged Pipelines and Equipment Coatings

Selection of Coverage for the coatings Specifications of the pipelines covering the 24" x 6.7km BBOU – ELEP BL and the 36" x 31.5 km ELEP – AKAB TL are:

- Surface Preparation
- Application of Primer
- Coating Repairs
- Materials
- Storage
- Inspection and Testing

16.6. Basis for Design

Risk of external corrosion can be mitigated by applying anti corrosion coating on line pipes to be buried, forming a barrier between the steel and the environment. Application of good coating reduces the C.P current requirement as both {Coating and CP} in conjunction provide external protection.

16.7. Design Requirements

The selection of coating material is based on

- The selection of materials that must meet the accepted standard from HSE view point.
- The material should provide high level of protection for the entire life of the pipeline.
- The adhesion between the steel and coating shall be effective throughout the life of the pipeline, with minimal disbondment of the coating.
- Durability of the resistance to chemical, physical and biological degradations.
- Maximum and minimum temperatures.
- Strength and impact resistance.
- Ease of repairs of the damaged coats.
- Compatibility with concrete weight.
- Acceptable unit price.
- Ease of application.
- Low permeability to water and salt.
- Permeability of oxygen should be low

16.8. Coating Materials Information

Table 15.1. Coating Materials Information

S/N	Materials	Operating Temperatures	Disbondment (Life Time)	Ease Of Application	HSE	Comments
1	Asphalt	X	X	√	X	
2	CTE	X	X	√	X	
3	Polyethylene Wraps	X	XX	√√	√	
4	Epoxy/PE/PP	√√	√√	√	√	Selected
5	FBE	√√	√√	√	√	

Legend: XX-Very Bad performance: X-unaccepted performance: √-Accepted performance: √√-Very Good performance

16.9. Selected Coat

Coating Material

On the basis of the analysis above, the selection of suitable coating material for this project should be cost effective and also depend on the operating temperature of between -35°C and 120°C. Hence, the coating even at higher temperature will favour the use of 3-layer epoxy polyethylene/ polypropylene in accordance with NACE specification 06053/ISO 21809-3.

Consequently, the Coating Materials Composition are:

1st Layer epoxy

2nd Layer Adhesive

3rd Layer PP top coat

It should be noted that the qualifications of coatings will comprise:

- Batch Certification
- Coating System Qualifications General
- Procedure Qualification
- Inspection and Test Plan
- Visual Appearance
- Coating Thickness, - see below table 15.2 on minimum recommended thickness for coating

Table 15.2. Coating thickness

Design Parameter	Unit						
Pipe size	Inch	10	12	16	20	24	28
Coating Thickness	mm	2.0	2.9	2.9	2.9	2.9	2.9

Other relevant items are:

- Holiday detection
- Impact Resistance
- Indentation Resistance
- Elongation
- Thermal Ageing
- Hot Water Resistance
- C.P Disbonding
- Degree of Cure of FBE Primer.

16.10. Coating Materials for Bend

Bends are factory coated using 3LPP as it relates to the pipeline

16.11. Coating Materials for Field Joint

The recommended field joint is the shrinkage sleeves type. However, the contractor may come with another system. The procedure application must be qualified and also approved by BFPCL before use.

16.12. Coating Materials Repair Works

The Contractor should provide the procedure of the repair methods and also the material for prequalification exercise before approval for use.

16.13. Surface Preparation

The line pipe will be prepared to SA 2.5 ISO 8501.4 (NACE No. 2) standards by blasting. It should be noted that for duplex SS the grit chips should be such that it does not leave any free Fe on the surface. A solution may be the use of Al grit chips.

16.14. Quality Assurance and Quality Controls

The QA, QC practice shall ensure that the coating procedures and specifications are met and should be in accordance with NACE specifications, standards and codes.

Key to achieving the goals will involve the following:

- Use of trained inspectors to the professional NACE standards
- QC inspectors should have proper equipment for testing and measurements. This should be approved by the BFPCL
- The inspector shall provide daily inspection records/reports.

16.15. Splash Zone or Tidal Level Coating Requirements:

Normally, splash zones coating requirements are associated with offshore structures but as this environment is close the Atlantic Ocean it is necessary for us to consider this zone to have very aggressive corrosion at the considered environment. It envisaged that at HHW water tide the

area could be flooded, swampy with running water. The coating selection should be such that have the ability to resist tear, wear and abrasion.

The splash zone of marine structure is exposed to alternate wetting and drying on daily basis hence, CP will not be effective at this zone and corrosion of the exposed metallic structures is expected to be accelerated. Consequently, the need for special anti-corrosion treatment shall be desired.

The main requirements that shall be considered for the treatment are:

- Bondability (ok with pipeline coat but not ok for piles etc)
- Resistance to brackish water
- Resistance to oxygen and ozone
- Resistance to tear, strength and abrasives

16.16. Splashzone Coating Materials Selection

It is always a challenge to select splashzone materials in swampy / flooded areas. From the Table 3 below the choice of coating materials is **splashTRON vulcanized elastomer** (highlighted item).

Table 15.3. Splashzone Coating Materials Selection

S/n	Material	Mechn. Prop. (Tear, Wear Abrasion)	Chem. Prop.	Marine growth resist.	Yard appl.	In-situ appl.	Ease of MTC.	Cost of matl.	Cost of appl.	Life time anti. Corr. Prot.
1	Splashzone 2 part compound	X	√	X	√	√√	√√	√√√	√√√	X
2	Neoprene Rubber	√	√	√	√√	X	√	√√	√√	√
3	Chlorinated Rubber	√	√	√	√√	√	√	√	√	√
4	Ethylene Propylene	√	√	√	√	X	√√	√	√	√
5	SplashTRON – Vulcanized Elastomer	√√√	√√√	√√√	√√√	√√√	√√√	√√	√√√	√√√
6	3-layer Epoxy/EP/PP	√	√	√	√	X	√	√	√√	√√
7	A788- 2-part compound	X	√	√	√√	√√	√	√√	√	X
8	Epoxy mastic materials	√	√	√	√√	X	√√	√√√	√√	√√

LEGEND: X = Bad, √ = Satisfactory, √√ = Very good, √√√ = Outstanding

For coverage of metals at this zone, it has excellent mechanical, chemical (in Brackish water) resistance and anti-corrosive qualities with anti-marine growth properties for pipings piles and

other structures etc.

Again, the pipeline or linepipe coating (3-layer Epoxy/EP/PP) is good enough at this zone.

16.17. Specifications

- The zone is established 1m below and above the tidal zone (HHW, LLW) – the insulation flange points and the CP lugs should be above this points so as not to leak corrosion protection current into other structures above this level.
- SplashTRON code 2216 application is obtained by applying 1/8" 2-part adhesive on the acetone cleaned surface of 1/4" thick split sleeve. This would be adhered into an already blasted metal surface to SSPC-SP5 or NACE no 1 surface preparation standards.
- The ratio of the mixture of the part A (gray) and the part B (white) is 3:2 by volume respectively.
- All other information on the materials application are classified by the manufacturer.

16.18. Quality Assurance and Quality Controls

The quality assurance shall be based on ISO 9001.

16.19. Marine Atmospheric Corrosion Protection

16.19.1.General

For the purpose of this marine atmospheric corrosion consideration for painting is any thin material applied as a liquid or powder which on solidification is firmly and continuously attached to the structures which it is designed to protect and prevent contact with the atmospheric environment as waterproof. The other application of paints is for aesthetic appearance and color. Certain parameters are used to achieve the selection of paints and these are incorporated into these specification document.

Wood, concrete and plastics and others are given limited consideration and reviewed against their deterioration principles. Historically, paints have been around for a long time and sometime the selection can be challenging as the manufacturers are many and varied. Hence, any selected manufacturer must be quality checked and approved by the principal to meet the standard requirements.

16.19.2.Facilities Coverage:

All facilities and equipment above the splashzone levels include:

- Manifolds
- Slug Catcher
- Valves
- Piping

- Launchers and Receivers
- Other above structures and equipment
- Metals and non-Metal systems (plastics, wood, concrete and other materials of construction).

16.19.3. Significant Parameters for Paints Selection:

- Temperature Range
 - Cryogenic conditions to -35°C (not considered)
 - From -35°C to 120°C (area of concentration)
 - Above 120°C (in few cases but rich zinc primer shall not be used)
- Preference of yard coatings for primer and mid-coat and top coat will be done along with the construction.
- Humidity (Relative Humidity below 80%)
- Paint thickness (wet or dry film thickness)
- Painting systems of each of the facility shall be provided by the same manufacturer as certified.
- The manufacturer or vendor to be approved by principal.
- Method of application
- Inspection and tests
- Daily reports
- Manufacturer paint shelf-life must be indicated.

16.20. Specifications:

The selection of paints for application at the atmospheric marine environment involves some compromise. The main considerations are:

- Good Paints material or system
- Paint applicator to do a proper job
- The application procedure fully incorporating the manufacturer instructions

Hence, the paint manufacturer and applicator shall be quality checked and approved by the principal.

Accordingly the following are specified:

- The metallic structures
 - The metallic substrate shall be cleaned, degreased and descaled (ISO/NACE)
 - Abrasive blast cleaning of the surface of the metal to SSPC – SP – 10 or NACE No.2 standards (used aluminium oxide as grit for stainless steel)
 - The painting system advised shall be for 120°C and below and the relative humidity to be below 80%.
One coat of rich zinc Epoxy primer (DFT 50 Microns).

One coat of micaceous iron oxide high built epoxy (DFT 125 Microns).

One of polyurethane paint (DFT 125 Microns).

External coating and tape sealing of flange assemblies at this zone require one coat of zinc phosphate epoxy primer.

Some paintings are to be done for aesthetic reasons which will include plastic, wood, concrete etc. The applicator shall provide procedures that must be agreed by the principles.

17. Attachments

17.1. Materials & Corrosion Datasheets:



17.2. Corrosion & Material Selection Drawings

